Haochen Luo (Chester)

Curriculum Vitae

chester.luo02@gmail.com — LinkedIn — GitHub

EDUCATION

Doctor of Philosophy in Computer Science
The University of Queensland, January 2027 – December 2030
Supervisor: Prof. Ruihong Qiu
Master of Philosophy in Computer Science
The City University of Hong Kong, January 2025 – December 2026
Supervisor: Prof. Chen Liu
Bachelor of Management in Information Management and Information Systems
Northeastern University (China), September 2020 – June 2024

SKILLS

Programming: Python, C++, SQL, CUDA
ML & Statistics: PyTorch, Machine Learning, Time-Series Modeling, Optimization
Quantitative Finance: Options, Derivatives Pricing, Volatility Modeling, Portfolio Optimization

INDUSTRY EXPERIENCE

Intern, Algorithm Engineer
September 2026 – December 2026
Hithink RoyalFlush, Hangzhou, China
Contributed to the development of a finance-specific foundation model, with a focus on strengthening its options-related capabilities. Built evaluation benchmarks covering option pricing, strategy construction, risk reasoning, and market understanding; developed data synthesis and quality-control pipelines for domain-specific instruction and reasoning data; and participated in model training, fine-tuning, and iterative evaluation to improve financial reasoning and option-related performance.
Intern, Machine Learning Algorithm Engineer
May 2024 – January 2025
DJI, Shenzhen, China
Contributed to DJI's internal AutoML framework, rewriting and optimizing PyTorch operators and designing hardware-aware search algorithms.
Developed a multimodal low-light data pipeline combining visual content, camera sensor signals, and imaging metadata for automated data curation, scene characterization, and hard-example mining.
Optimized low-light enhancement models for the Osmo Action 5 Pro, incorporating multimodal contextual cues to improve generalization across diverse scenes and lighting conditions.
Intern, Algorithm Engineer
October 2023 – May 2024
ASML Brion Technologies, Shenzhen, China
Developed an enterprise RAG system for computational lithography and defect analysis, integrating internal technical documentation, defect records, and structured databases to support domain-specific knowledge retrieval and automated analysis; built LLM tools to streamline engineers' defect review and troubleshooting workflows.
Built high-performance data processing infrastructure for computational lithography and defect analysis, optimizing multi-threaded C++ pipelines and large-scale data I/O to support downstream ML workloads; developed Python-based visualization tools for efficient model and defect analysis.
Intern, AI Algorithm Engineer
July 2023 – October 2023
XGrids Innovation, Shenzhen, China
Developed a custom sparse convolution operator in CUDA for point cloud processing on NVIDIA Jetson embedded devices, applying kernel-level optimizations and memory access tuning to accelerate parallel computing and reduce GPU latency; integrated the operator into point cloud segmentation networks, with ONNX export and deployment support for a C++ Qt interface.

RESEARCH EXPERIENCE

LLM Applications in Quantitative Finance

- Conducted research on applying large language models to quantitative finance, focusing on alpha factor mining, signal discovery, and structured trading strategy generation.
- Developed LLM-guided frameworks for generating interpretable financial factors, leveraging prompt design and evolutionary search to improve performance and diversity.
- Designed structured intermediate representations (e.g., domain-specific languages) to bridge natural language reasoning and executable financial strategies.

Privacy and Robustness of Large Language Models

- Studied machine unlearning and privacy-preserving techniques, including differential privacy, for responsible deployment of large models.
- Investigated adversarial vulnerabilities in vision-language models and explored defense strategies to enhance model robustness and reliability.

SELECTED PREPRINTS

Haochen Luo, Sifan Lin, Xiaolong Luo, An Binh Minh, Yuan Zhang, and Chen Liu. "Arbitrage-Free Implied Volatility Surface Reconstruction via Additive Martingales." *arXiv preprint arXiv:2610.25671, 2026.*

Ding Chen, **Haochen Luo**, and Chen Liu. "CommuteProp: Decoupled Training for Communication Bound Split LLM Fine-Tuning." *arXiv preprint arXiv:2610.05105, 2026.*

SELECTED PUBLICATIONS

Haochen Luo, Yifan Li, An Binh Minh, Xiaolong Luo, Zhengzhao Lai, Yuan Zhang, and Chen Liu. "LiveOption: Evaluating LLM Agents in Structured Option Trading with Nonlinear Payoffs." *Neural Information Processing Systems (NeurIPS) 2026, Evaluations and Datasets Track.*

Haochen Luo, Yifan Li, Ho Tin Ko, An Binh Minh, Junjie Xu, Tang Pok Hin, Wang Chak WONG, Gao Yuan, Zhengzhao Lai, Yuan Zhang, and Chen Liu. "QFinZero: A Unified Financial Toolchain for LLM-Based Trading Agents." *Annual Meeting of the Association for Computational Linguistics (ACL) 2026, Demo Track.*

Haochen Luo, Ho Tin Ko, Jiandong Chen, David Sun, Yuan Zhang, and Chen Liu. "AlphaBench: Benchmarking Large Language Models in Formulaic Alpha Factor Mining." *International Conference on Learning Representations (ICLR) 2026.*

Xuyang Zhong, **Haochen Luo**, and Chen Liu. "DualOptim: Enhancing Efficacy and Stability in Machine Unlearning with Dual Optimizers." *Neural Information Processing Systems (NeurIPS) 2025.*

SERVICE & PROFESSIONAL DEVELOPMENT

Peer reviewer for the Association for the Advancement of Artificial Intelligence (AAAI), International Conference on Machine Learning (ICML), Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR), Conference on Language Modeling (CoLM), and Transactions on Machine Learning Research (TMLR).